

State of (state name) Operator Certification Program CY20XX Annual Report to the US Environmental Protection Agency

[month day, year]

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This model report does not add nor modify any existing requirements which are found in the Final Guidelines and nine baseline standards. Reference Documents for this model report are as follows:

[EPA Desk Guide for Reviewing State Operator Certification Program Annual Reports²](#)

[Final Guidelines with nine baseline standards³](#)

² [EPA Desk Guide for Reviewing State Operator Certification Program Annual Reports](#)
[epa.gov/epa-608/epa_desk_guide_for_reviewing_state_opcert_program_annual_reports.pdf](#)

³ [Final Guidelines with nine baseline standards](#)
[epa.gov/epa-608/final-guidelines-with-nine-baseline-standards/](#)

Introduction

In 1999, the U.S. Environmental Protection Agency (EPA) issued operator certification program guidelines specifying minimum standards for certification and recertification for the operators of community and non-transient non-community water systems (CWS and NTNCWS). The goal of the operator certification program is to protect human health by ensuring that skilled professionals are overseeing the treatment and distribution of safe drinking water. Operator certification is a crucial step in promoting compliance with the Safe Drinking Water Act (SDWA), and oversight of the program ensures that these steps are taken. Failure to adopt and implement an approved program requires EPA to withhold 20 percent of the SDWA DWSRF funds that the State would otherwise be entitled to receive.

The State of (state name)'s Operator Certification Program (program) was approved as consistent with the *Final Guidelines for the Certification and Recertification of the Operators of Community and Non-Transient Non-Community Public Water Systems*, 64 CFR 5916 (Final Guidelines) on (enter date of your state approval).

This is State (state name)'s annual program report to EPA for calendar year 20XX (CY20XX). In accordance with the Final Guidelines, this report addresses the status and continued implementation of State (state name)'s program for the nine Baseline Standards for both CWS and NTNCWS. The nine program components are:

1. Authorization
2. Classification of systems, facilities and operators
3. Operator qualifications
4. Enforcement
5. Certification renewal
6. Resources needed to implement the program.
7. Recertification
8. Stakeholder involvement
9. Program review

The State (state name) Operator Certification Program ensures no backsliding and no policy change occurred in CY20## in the operator certification program; the program remains the same since last year.

Public Health

State (state name) Operator Certification Program, in conjunction with other State (state name) EPA programs, ensure that customers of water systems are provided with an adequate supply of safe, potable drinking water and that consumers are confident that their water is safe to drink. State (state name) Operator Certification Program ensures water [and wastewater] operators are appropriately trained, have an appropriate amount of experience and demonstrate a knowledge and understanding of the public health reasons for drinking water standards.

Authorization

Example authorities below; insert your state's program, department, statutory, and regulatory authorities:

The Water [and Wastewater] Facility Operators Certification Board (board) is a section of the Division of Administration within the State (state name) Department of Public Health and Environment. Article 9 of Title 25, State (state name) Revised Statutes (C.R.S.), requires every water [and wastewater] facility to be under the supervision of a certified operator holding a certificate equal to or greater than the classification of the facility. The board is authorized by 25-9-101 et. seq., C.R.S., to promulgate Water [and Wastewater] Facility Operators Certification Requirements, Regulation 100, 5 CCR 1003-2, (Regulation 100).

The State (state name) Department of Public Health and Environment, Water Quality Control Division (division) has been delegated responsibility for the oversight of public water systems in State (state name) to ensure compliance with and enforcement of the provisions of Regulation 100, the Safe Drinking Water Act, Regulation 11-State (state name) Primary Drinking Water Regulations and other pertinent regulations.

If changes were made to the regulation or statute:

The state must include a copy of a new Attorney General's (or delegated counsel's) Certification, amend their program description (including discussion of meeting the antibacksliding provision), and include a copy of the new regulations or statutes.

Classification of Systems, Facilities, and Operators

Public water systems

The division is responsible for applying board regulation and direction to classify all public water system treatment and distribution system facilities in accordance with the provisions of Regulation 100. Treatment classification is based on specific design features that include treatment processes and their complexity, source water type and design capacity. Distribution systems are classified by the population served. Each classification has a specific certification required for the operator in charge. Water treatment facilities are classified as Class D, Class C, Class B, or Class A. Class A is the highest level of classification. Water distribution systems are classified as Class 1, Class 2, Class 3, or Class 4. Class 4 is the highest level of classification. Class S systems serve a population of less than 3,300 and combines treatment Class D and distribution Class 1 with basic treatment processes. Table 1 represents these classifications.

Table 1 – Total Number of CWS and NTNCWS in Each Classification

System Classifications	CWS	NTNCWS
Water Treatment		
Class A	459	20
Class B	50	11
Class C	20	7
Class D	1	11
Class S	80	21
Water Distribution		
Class 4	212	21
Class 3	80	15
Class 2	20	10
Class 1	15	8

Certified operator

Regulation 100 requires every water [and wastewater] facility to be operated under the supervision of a certified operator in responsible charge who holds a certificate equal to or greater than the classification of the facility. The regulation also reserves all process control and system integrity decisions to the certified operator in responsible charge or another operator holding a certificate equal to or greater than the classification of the facility in accordance with a written operating plan. The division also continues to utilize education, tracking, and enforcement measures to ensure that all CWS and NTNCWS operators are certified. The certified operator in responsible charge or another operator who holds a certificate equal to or greater than the classification of the facility must be available at all times during operating hours of a water treatment or distribution facility. Table 2 below provides a breakdown of the number of certified operators for each level of classification.

Table 2 – Number of Certified Operators by Classification

Classification	Number of Water Systems	Number of Certified Operators
Class A	459	845

Class B	50	41
Class C	20	29
Class D	1	10
Class S	80	107
Class 4	212	282
Class 3	80	95
Class 2	20	28
Class 1	15	10

Contract operations

State (state name) has guidelines that will allow water systems to achieve compliance with the state's regulation by contracting with an operator that is not a direct employee of the system. State (state name) must approve all contracts that intend to achieve compliance with the certification regulation and must receive an actual copy of the contract. The premise of a contract operator is they must be in direct charge of the water system; however, they can have other people work under their supervision. They do not necessarily have to perform day-to-day operating task including sampling. One of the key areas examined by State (state name) is the number of on-site visits per week to the water system and the duration of these visits.

Compliance

Compliance with operator certification requirements is a State (state name) Safe Drinking Water Program priority. Non-compliance is a result of one of the following circumstances:

1. A certified operator has not been designated by the system's owner.
2. The operator's certificate has expired.
3. The operator's certificate is not at the correct level for the facility they are operating.

As of [month year], 13 systems do not have a certified operator. With measures in place through Regulation 100, 12 of these systems have temporary operators in place.

Recommendations: *Include and discuss activities related to workforce/recruiting, such as number of trainee (Operator In Training or OIT) designations for operators working under an operator in responsible charge (OIRC). Include names of systems where classifications were changed (and reasons) within the reporting period. It is also recommended to not just state, but briefly describe policies/procedures in place that ensure operators are certified and available each operating shift including monitoring of the system when a properly certified operator is not physically on-site (e.g., standard operating practices (SOPs), Supervisory Control and Data Acquisition (SCADA)).*

Operator Qualifications

Operator exam

State (state name) requires that each operator must pass an exam appropriate for the classification to which they are applying for. There are separate treatment and distribution exams and all exam questions are validated prior to inclusion in current exams. State (state name) also grants a Class S certificate that combines treatment Class D and distribution Class 1 operator certificates designed specifically for public water systems serving a population of less than 3,300 with basic treatment processes. Many operators hold multiple certificates.

Exam pass rates

Successfully passing exams is vital to maintain a steady workforce of certified operators. Low pass rates at the higher levels continues to be an area of concern. In 20XX, the program began requiring sequential testing. Sequential testing requires the applicant to hold a valid certificate at the level immediately below the level of the exam they apply to take.

The 20XX average pass rates for water treatment and distribution written exams are 43.5 percent and 61.5 percent for electronic exams (shown in Tables 3 and 4). Because the questions are identical, regardless of the format, the program attributes the difference in pass rates between written and electronic exams to several factors. Traditional paper and pencil exams are taken in a room with as many as 400 people. The entire exam booklet is in front of the examinee, which can be overwhelming and distracting. By comparison, electronic exams are administered in a room with less than 24 people taking the exam at a time. The atmosphere is quiet, more relaxed and questions are presented one at a time, which reduces distractions and allows the examinee to focus on the question in front of them.

Table 3 – Written Exam Pass Rates

Level	# of Exams	% Pass	Level	# of Exams	% Pass
WA	43	14	D4	47	17
WB	51	61	D3	68	22
WC	45	78	D2	108	44
WD	113	60	D1	208	26
WS	49	88	-	-	-
Total Exams or Average Passing %	301	60	Total	431	27

Table 4 – Electronic Exam Pass Rates

Level	# of Exams	% Pass	Level	# of Exams	% Pass
WA	79	38	D4	92	33
WB	69	61	D3	88	49
WC	69	75	D2	125	57
WD	121	78	D1	187	73
WS	27	96	-	-	-
Total	365	70	Total	492	53

The average pass rate for all water treatment exams is 65 percent. The average pass rate for distribution exams is 40 percent. The overall pass rate for water treatment and distribution exams is 52.5 percent. This represents a drop from an average passing rate for all exams of 61 percent in the previous year. The program has not made any changes that would explain the percentage drop in exam pass rates for the reporting year. The 20XX average pass rate is in keeping with the historic program exam pass rates. In evaluating exam results, changes to exam format and content were discussed among the Department, subject matter experts, and a sample of certified operators.

Training and education requirements for certification

Applicants for certification are required to provide proof of education, such as a copy of high school, general equivalency, or a college diploma.

Applicants must also provide proof of all initial and/or continuing education training by submitting training certificates. The Department verifies this against class rosters and examination scores submitted by the training providers. Furthermore, experience, including job duties and responsibilities, is verified by the employer. The employer will not conflate training requirements with education requirements and vice versa.

***Recommendation:** Include in the report a brief discussion of the state's reciprocity approval process, if applicable, and the number of new operators based on reciprocity.*

Grandparenting

This option was only available to water distribution operators and was valid for two years. The grandparenting provision allowed an operator who was certified before January 30, 2001 under the voluntary program administered by the State (state name) Water [and Wastewater] Collection Systems Certification Council, Inc. to continue operating for two years.

All operators were required to obtain a standard or a restricted certificate no later than January 30, 2003. A standard certificate required qualifying for and passing a standardized exam. The restricted certificate was issued to an operator who was certified under the voluntary program and was renewable upon payment of fees and having obtained the required training units. Restricted certificates were valid as long as the operator continued employment at the same facility where they worked on January 30, 2001 and the facility classification remained the same.

As of [month year], there are 99 operators holding 125 restricted certificates. All others granted a restricted certificate have either allowed the certificate to expire or have successfully passed a certification exam and no longer rely on a restricted certificate.

***Recommendation:** Include in the state report discussion on any Operator Certification-Capacity Development partnership activities. See example below.*

Operator Certification-Capacity Development partnership

The Local Assistance Unit was developed by the State (state name) Safe Drinking Water Program. This unit works in partnership with drinking water systems, assistance partners and division staff from various program areas. The partnership reaches operators and system owners on a grassroots level to improve compliance, enhance water quality, better position a system managerially, financially, and technically for the future and to protect public health. The Local Assistance Unit's purpose is to provide technical, management and financial assistance one-on-one and through group trainings to public water systems so they can strengthen their ability to supply safe drinking water to the public and to protect their source water.

The operator certification staff person serves as the liaison to the Water [and Wastewater] Facility Operators Certification Board (WWFOCB), provides outreach and compliance assistance to operators and system owners and is in the Local Assistance Unit. This ensures alignment with capacity development efforts including training, certification challenges and improved assistance to facility owners.

The division has increased outreach to operators in an effort to address the identified areas of concern:

- ☐ The division sponsors no or low-cost trainings offered in multiple sites across the state with a focus on operations, math, and regulations.
- ☐ In 20XX, division coaching staff provided 186 on-site, individual operator training sessions at no cost and class room style training sessions and presentations at other schools and conferences.
- ☐ Division compliance, operator certification and engineering staff regularly present at conferences, seminars and other training venues throughout the state on various regulatory compliance topics.
- ☐ Operator certification staff participates in the Funny Mountain Section AWWA small systems education committee which provides training targeting small system operators.
- ☐ The Local Assistance Unit provides support for State (state name) Water [and Wastewater] Agency Response Network (WARN) and works with them to provide systems and operators with emergency preparedness and response training.

Enforcement

Example authorities below; insert your state's enforcement and compliance authorities:

Facility enforcement

Enforcement authority under 25-9-110(3), C.R.S. is granted to the department to issue and enforce administrative orders and to impose civil penalties against facility owners for failure to operate their systems under the supervision of a certified operator. State (state name) references database information and sanitary surveys to determine compliance and non-compliance at facilities.

In 20XX, there was one formal enforcement action (also called Notice of Violation) against a public water system for violating the certified operator requirements. Twelve other systems were issued a notice of noncompliance but through Regulation 100, these systems have temporary operators in place.

Operator disciplinary action

The board has authority under 25-9-104(6), C.R.S. to establish criteria for the discipline or reprimand of an operator. Section 100.23.3 grants the division authority to investigate instances of possible misconduct by certified operators. In 20XX, there were nine new disciplinary action investigations in 20XX. One system was investigated due to system non-compliance with the Operator Certification regulations. Two System Investigations were dismissed for lacking sufficient evidence of misconduct. Four investigations resulted in the issuance of informal letters of reprimand from the division. One resulted in the board revoking both the operator's water treatment and water distribution certificates, which the state reserves the right to do.

Table 5 demonstrates licenses revoked and suspended in 20XX due to the aforementioned enforcement procedures.

Table 5 – License Revocations and Suspensions at the End of the 20XX Reporting Period

Licenses Revoked	Licenses in the Process of Revocation	Licenses Suspended	Licenses in the Process of Suspension
0	1	3	1

Table 6 provides the five-phase enforcement process and the action to be taken by State (state name) staff if a system is found to lack a qualified operator.

Table 6 – Enforcement Steps

Type of Violation (No Qualified Operator)	Phase	Number of Days Following the End of the Compliance Period	Action to be Taken
PWS determined to not have a qualified operator	Phase 1	Thirty days	Issue notice of non-compliance.
PWS determined to not have a qualified operator	Phase 2	Sixty days	Issue notice of non-compliance, with warning of administrative order. Phone call confirming administrative order.
PWS determined to not have a qualified operator	Phase 3	Ninety days	Issue notice of non-compliance, issue administrative order and give 60 days to comply, with warning of penalty.
PWS determined to not have a qualified operator	Phase 4 & 5	One hundred and fifty days	Assess administrative penalty for violation with escalating penalty.

Certificate Renewal

Certificate renewal

Regulation 100 requires all active operators receive on-going professional development in order to renew certificates every three years. As an incentive to maintain certificates in current status, a late fee of \$50 is charged to an operator who submits the renewal application after the certificate's expiration date.

During CYXX, 1,160 renewal applications were approved. The certificate renewal breakdowns are as follows: Class A = 802, Class B = 20, Class C = 15, Class D = 2, Class S = 57, Class 4 = 190, Class 3 = 60, Class 2 = 12, and Class 1 = 2. During the fiscal year, 137 renewal applications were rejected, some of which were later approved during the same year.

Professional development

Training courses are offered throughout the year to fulfill the professional development requirement for renewal and must be approved by the board or its contractor. See Table 7 below for the number of approved operator training courses. Operators who fail to renew or qualify for renewal within two years of the expiration of their certificate must complete the recertification process. Details on recertification are on page 19.

Individual operators are responsible for tracking and reporting training units for certification renewal. In 20XX, 30 operators were required to take additional training based on state evaluations and depending on the system of which the operator is in responsible charge. All operators must acquire the necessary amounts and types of state approved training. Grandparented operators may require additional certification, dependent on how compliant said system has been, and operator experience and education. Evaluated on a case-by-case basis, 23 grandparented operators were required to complete additional training, out of the 45 with active licenses.

Table 7 – Approved Operator Training Courses

	20XX	20XX	20XX	20XX	20XX
Initial Courses	66	48	48	46	56
Renewal Courses	1,100	1,177	1,056	998	1,064
Total Courses	1,166	1,225	1,148	1,044	1,120

Certificates of completion and transcripts are reviewed to determine fulfillment of the professional development requirements. In 20XX, more than 500 courses approved for operator professional development were offered. Many of these classes were held multiple times within the year. The price for each course varies from no charge to several hundred dollars for multiple day training.

State (state name) does not maintain a clearinghouse that tracks the number of classes taught by an instructor or institution, the number of attendees at each training class or the cost to the

student for each course. Courses approved for training units are not ranked by certification level.

Trainers submit applications for training units which are approved by a committee of volunteer subject matter experts who review all applications and assign the training units. Upon approval, the course is entered into the database of approved training courses. A list of approved courses is available online at [www.state\(state name\).gov/operatorcertificationprogram/professionaldevelopment](http://www.state(state name).gov/operatorcertificationprogram/professionaldevelopment).

Resources Needed to Implement the Program

State (state name) provides sufficient resources to contract the operator certification and examination aspect of the program to a non-profit organization comprised of subject matter experts who volunteer their time and expertise. The contractor subcontracts daily operations to Total Events and Management Services, Inc. (TEAMS). TEAMS manages the operator certification program office, which oversees the daily operation of the operator certification portion of the program.

Income to support the operator certification program office originates from fees charged to applicants and operators as authorized by statute and Regulation 100. These fees cover the costs of management for this portion of the program. In 20XX, subject matter experts volunteered nearly 3,400 hours reviewing applications and training courses, proctoring exams and completing other contractual activities. This is equivalent to approximately one and a half full-time employees.

Operator certification fees do not fund agency staff. All agency staff funding is provided through state general funds, general program fees and various federal grants. The division has one full-time employee dedicated to operator certification who serves as staff to the board and as the division liaison to the board. In addition, compliance with the requirements of Regulation 100 is performed by the drinking water compliance and enforcement staff and compliance is checked on-site by field services section staff during sanitary surveys. State (state name) program staff receive training on evaluation of compliance with operator certification requirements as an element of their sanitary survey training. For example, staff will ensure the operator certifications have not expired and ensure the facility classification that was done at the time of design review by engineering section staff still applies. Noncompliance with operator certification will be identified as a significant deficiency. State (state name) strongly encourages the system's certified operator to participate in sanitary surveys. The division's data system, management and administrative teams provide necessary support for the program as well. The board's administrator and staff represent approximately 0.2 full-time employees. In total there are over 65 division staff members with responsibilities related to the certified operator program. Under the current organization, State (state name) has sufficient resources to implement its operator certification program now and into the future.

The PWSS grants and DWSRF set-asides remain the same and continue to sustain the program.

Recertification

Certificates are valid for three years from the date of issue. Once an operator's certificate has expired, they are no longer certified. A certificate may be restored, through renewal, for up to two years after the expiration date. A certificate is automatically revoked after two years if it is not renewed. Subsequently, if the person desires to be recertified they are treated as a new applicant for certification and must sequentially re-test.

The Department requires a full review and examination of the operator's experience, training and validated examination as contained in Subpart 5-4.3(b) and 5-4.5, as describe above under initial certification. Although no operators were recertified during this reporting period, if there had been, the number of operators would be listed in this section of the report.

Stakeholder Involvement

Rulemaking

In November 20XX, the division began the regulation revision process. Approximately 8,000 people were invited to participate in the stakeholder process. Those invited included certified operators, system owners, and other interested parties. After the conclusion of this stakeholder consultation process, the division held a public hearing to consider the proposed amendments to the regulation on [month day, year]. The WWFOCB promulgated changes to Regulation 100 at that time.

The Office of Legislative Legal Services, which conducts legal reviews of all newly-promulgated agency rules, expressed concern after the [month year] adoption of section 100.15.3 that the language describing "professional manner" was too vague to meet the requirements of the State (state name) Administrative Procedures Act, 24-4-101 et. seq., (state name). To address these concerns, the board initiated a brief stakeholder consultation process and then held a rulemaking hearing on [month day, year] and adopted changes to section 100.15.3 to delete the concept of "professional manner," and instead to specifically prohibit certain types of behavior when a certified operator is acting in a professional capacity. These prohibited behaviors are related to a certified operator's demeanor while communicating/interacting with the public, the regulated community, and regulators. The board similarly revised section 100.20.1(j) to clarify that certified operators who exhibit such behavior are subject to disciplinary action by the board.

The current Regulation 100 is included as Appendix A of this report along with a summary of changes made as Appendix B. *(Note: The Appendices are not part of this model report but would be included in an actual report).*

Recommendation: *Host a stakeholder board or advisory committee, instead of simply using public comment processes.*

Drinking water group meetings

The Drinking Water Group held six meetings in 20XX; all meetings are open to the public and participation is encouraged. The April meeting was held in conjunction with the annual State (state name) Rural Water Association conference. There were approximately 100 in attendance at the [month year] meeting.

Water newsletter

The Safe Drinking Water Program publishes a quarterly newsletter that is distributed to approximately 3,000 system owners, operators, and others. The newsletter is also published and available on the division website. Each newsletter contains at least one article specifically focusing on operator certification topics of interest and any changes to regulation or board policy. Conduct an internet search for “(state name) *EPA Water*” to find electronic copies of the Water newsletter.

Program Review

The division conducts periodic reviews of its processes, procedures and data management. Additionally, regulations, training needs and exam items are reviewed with the stakeholders regularly. See Appendix C for meeting minutes from the program review meeting. *(Note: The Appendix is not part of this model report, but would be included in an actual report).* Each public water system is assigned to one compliance specialist. The compliance specialist oversees all compliance and enforcement activities related to a specific water system in regards to both drinking water regulations and operator certification regulations. This is a holistic approach that enhances communication between the division and the regulated community and has improved program efficiency and effectiveness.

The division continues to work toward developing a standardized approach to conducting both internal and external reviews.

***Recommendation:** Perform periodic internal reviews and occasional external/peer reviews. Examples of items to review include: regulations, exam items for relevancy and validity, compliance, enforcement, budget and staffing, training relevancy, training needs through examination performance, and data management system.*

Conclusion/Summary

The State (state name) Operator Certification Program is meeting the requirements for the nine Baseline Standards for CWS's and NTNCWS's in accordance with the Final Guidelines to deliver safe and reliable drinking water. State (state name) continues to support and enhance its data systems for improved tracking and management of certifications, water system information, operators' relationships with water system, and compliance with requirements.

Updates and Areas of Focus

The State (state name) Operator Certification Program continues to have concerns regarding replacement of retired certified operators. To address this concern, we are working with water [and wastewater] associations, towns, cities, schools, students, and water [and wastewater] utilities throughout the state to promote the profession. Our goal is to increase the awareness of the job and introduce water related courses into trade schools.

Recent events have raised State (state name) concerns about cybersecurity vulnerabilities in critical water infrastructure. State (state name) is currently using EPA's [Water Sector Cybersecurity Trainings](#) to start discussing the issue with the state's operators. The state has also started discussion with the state's technical assistance providers about adding information about assessing cybersecurity practices and developing an action to their portfolio.

Outreach and assistance along with training are important and ongoing aspects of the program to ensure system compliance. State (state name)'s recent focus on improving continuing

education programs has resulted in a substantial expansion of training opportunities for the state's operators. The online workshops have allowed the program to reach operators across the state, including in the more rural areas. Similarly, State (state name) also continues to prioritize at-risk systems needing additional assistance through the [Capacity Development Program](#), understanding that technical assistance for monitoring and reporting violations include operate compliance training.